Pinned chained-transfer and small-prime oracle

2026-11-11 · cryptanalysis · producer-run development checks

Standing: unsigned and unverified. The unchanged pre-existing BigUint arithmetic checks complete finite class sets. Both implementations use Cantor; the field-point counter shares the existing Fp2 lineage. Agreement is not signed independent validation of every layer. No benchmark or computational advantage is claimed. [pcc:unverified]

Source custody and execution scope

Current crypto pin:75a737b8366ec6c78ed8672886423fcbe69853a2. Seven runtime SHA-256 inputs include native Cargo/source files and the external unchanged src/prime_hyperelliptic/fp2.rs. The consumer enables small-prime and refuses source/reference drift before compiling. The reference files remain pinned to cryptanalysis68594c8874c75a0a6c04a3874adb1374088ca4f5.

check.sh passes native formatting, strict Clippy and25 tests, then reference formatting, strict Clippy and33 tests with1 existing benchmark ignored. Raw receipt cubic-pinned-02.txt; PCC-20261111-29610e remains unsigned. Earlier18db046,b280efc,aabae197 pin metadata and every raw receipt remain retained. No failed attempt was replaced. [pcc:unverified]

Exact four-field comparison

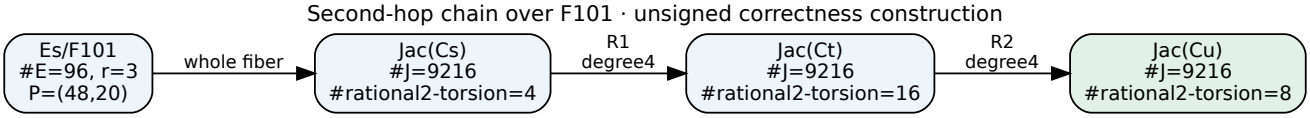
p	Each Jac order	Self-inverse source/target	Elliptic quotient orders
13	180	4 / 4	10 / 18
29	960	4 / 16	40 / 24
31	1008	16 / 4	28 / 36
101	9216	4 / 16	96 / 96

On all eight complete models, BigUint independently computes the monic chart change, paired additions, dual norms, self-inverse counts, Fp/Fp² Jacobian orders and elliptic quotient orders. The first13/29/101 fields were fixed before execution;31 was separately added as a nonresidue control and uses3 rather than2. [pcc:unverified]

Different rational2-torsion counts obstruct a base-field Jacobian isomorphism; equal counts at13 do not establish one. The source prime subgroups are5,5,7,3. The native backend validates odd primes≤4096 and caps enumeration at101. These tiny correctness fixtures do not establish large-field performance, cryptographic-size support or transfer between prime characteristics. [pcc:unverified]

The earlier r17 fixture retains10,404 source additions/inverses,289 target subgroup additions, complete Richelot-image translation checks and independent point-count orders. Its initial target is source-isomorphic and its complete elliptic endpoint matches a direct shortcut. [pcc:unverified]

Retained two-hop construction and limits



The predeclared successor second pairing has determinant67, base11 and third-model coefficients[38,9,29,62,86,17,8]. All9,216 target classes have BigUint additions/dual norms checked, independent order9,216 and self-inverse count8. The rational composite kernel has8 elements,4 self-inverse; do not confuse it with geometric degree16. Source P=(48,20),r3 reaches the canonical class $u(T)=T^2+86T+27$,$v(T)=34T+7$. Upstream native checks include trivial next-kernel/previous-dual intersection and reversed-chain/source norm[8]. [pcc:unverified]

The source global successor argument derives n R f∗ = tau phi[2] for its first elliptic projection in characteristic≠2,3,5, with tau(X,V)=(X/100+1/125,−V/1000). The actual base translation(1/125,0) is included. Seven cleared ring lemmas compile upstream; projective/Picard completion is a written argument, not a full Lean construction. It is not asserted as a collapse of the final Jacobian-valued two-hop map. [pcc:unverified]

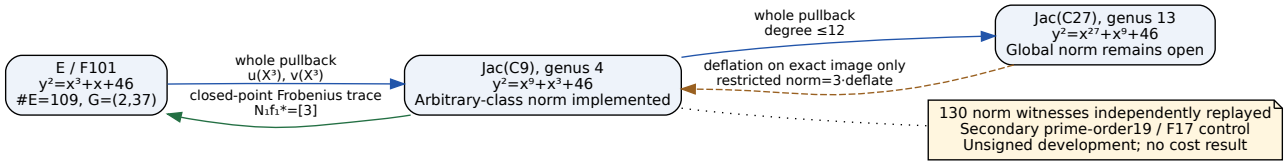
Reproduction and review

Check out crypto and cryptanalysis as siblings, with the exact pinned native and external Fp2 files, then run experiments/chained-transfer-reference/check.sh. Cargo.lock and source manifests freeze the dependencies. The focused push-main/manual workflow checks out that same runtime closure and runs both suites. Actual final-head CI is tracked in GitHub separately from local receipts. Prior focused run42986170752 succeeded on exactc6bc590; its raw Actions log is retained without treating it as signed PCC scientific standing. [pcc:unverified]

Crypto’s original PR3877 remains subject to its current-main ledger-prefix fork and historical archived-receipt gates. No history or verifier was rewritten. S3 checkpoint credentials, live IsoLab budget/capacity, KB/ecdb and the autoresearcher vault binding remain unavailable. No performance run has been launched. The queue is crypto issue3904; locked-atlas integration stays in its source handoff.

Primary sources: Galbraith corrected ch.8 §8.3,ch.10 §§10.3–10.4/10.6; Kunzweiler,DCC92(2024),DOI10.1007/s10623-024-01366-1,Propositions7–8. The new arithmetic covers bounded fixtures; independent signed validation, full geometric formalization and any computational advantage remain open.

Cubic arbitrary-class norm prerequisite



The existing Enge20261024 cover family already supplies C(d,s). This extension implements the bounded missing arbitrary-class norm on genus4 using unchanged BigUint polynomial/Cantor arithmetic. The pre-existing E109/F101 fixture has cubic covers of genera4 and13; whole pullback substitutes X^3 in u,v. Deflation recovers the exact source class on the coefficient image. [pcc:unverified]

For arbitrary canonical genus4 inputs, support u is completely factored through degree4. In each residue field Fp[X]/q the norm sums Frobenius conjugates of $(X^3,v(X))$, checks orbit closure/descent and retains support multiplicity. The map takes no scalar labels. Checked primes are5..101. The second norm remains restricted to the exact pullback image; no global arbitrary-genus13 norm is implemented. [pcc:unverified]

All109 source scalars pass both maps, norm[3] and composed restricted norm[9]. The fixed130 cases kA and kA+B,k=0..64 have closed support degrees1..4. A separate root crypto example reconstructs their sources, genus13 images and elliptic norms with pre-existing arithmetic, plus a distinct Frobenius-gcd irreducibility criterion. The producer norm is not imported. Both Jacobian engines use Cantor and the harnesses are producer-authored; PCC-2d5c08 is unsigned. [pcc:unverified]

The E19/F17 control checks361 ordered subgroup pairs, ramification and degree12 representatives. Four new tests and the full guarded25+33 suite pass. First compile/PATH failures remain retained separately. Four structural Lean lemmas compile upstream; Picard/finite-field geometry is not formalized. [pcc:unverified]

Odd full elliptic group order prevents rational linear/quadratic factors in $f(X^2+s)$, restricting the present rational-factor constructor. It does not exclude a Frobenius-stable unordered splitting into nonrational quadratics. Next: balanced genus2 arithmetic, then an actual non-split descent assessment. No genus increase, point recovery or source-group embedding alone supplies a DLP speedup. No benchmark or prime-field descent is claimed. [pcc:unverified]

Source-linked account and witnesses: crypto research/chained_jacobian_transfers_20261110/CUBIC_NORM_20261111.md. Primary geometry: Galbraith corrected ch.8§8.3, ch.10§§10.3–10.4. Existing Enge relation-yield and total-cost gaps remain open. Canonical transfer/leaderboard graphs remain unchanged: no signed finding or performance edge has been promoted.